

Joaquin E. Rivas Sánchez

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EDUCATION

University of Informatics Sciences

Feb 2022 — Jul 2025

Bachelor's in Computer Science and Engineering (CSE)

Havana, CU

- Cumulative GPA: 3.86/4.0
- Relevant coursework: Machine Learning, Database Systems and Optimization, Data Structures, Linear Algebra, Discrete Mathematics, Differential and Integral Calculus (single and multivariable).

Moscow State University of Geodesy and Cartography (MIIGAiK)

Oct 2024 — Jan 2025

Course: Modern Methods and Technologies for Spatial Data Processing

Moscow, RU

- Studied modern techniques for processing, analyzing, and managing spatial data using geodetic and GIS tools.
- Gained hands-on experience with advanced methods for acquisition, transformation, and accuracy evaluation of spatial data.
- Explored current technologies in remote sensing, digital cartography, and geospatial data workflows.
- Learned practical approaches for handling large-scale geospatial datasets and integrating them into mapping and analysis systems.

EXPERIENCE

Teaching Assistant – Machine Learning

Sep 2025 — Present

University of Informatics Sciences

Havana, CU

- Delivered practical sessions on machine learning workflows, Python, NumPy, pandas, and Scikit-Learn.
- Assisted the lead professor during lectures and collaborated on course planning and material preparation.
- Graded assignments, exams, and machine learning projects fairly and accurately.
- Provided one-on-one tutoring and academic support to students.
- Contributed to improving course exercises and updating the curriculum.

PROJECTS

RAG Chatbot (github.com/humankernel/rag-revamped)

Jan 2025 — Jun 2025

- Built a complete RAG pipeline (document segmentation, embeddings, vector search, context construction, LLM generation).
- Implemented modular components: loader, segmenter, embedder, retriever, aggregator, and generator.
- Integrated with OpenAI and vLLM models for flexible local or API-based inference.
- Developed evaluation tools for retrieval quality, latency, and model accuracy.
- Authored full system documentation with architecture diagrams and unit tests.

Chess Engine (github.com/humankernel/chess-engine)

Aug 2025 — Oct 2025

- Implemented a chess engine based on **bitboards** in C using a 64-bit board representation.
- Created attack tables for pawns, knights, and kings, including rank masks and edge conditions.
- Designed evaluation heuristics: material balance, pawn structure (isolated, doubled, blocked), and mobility.
- Implemented bit manipulation utilities: LSB extraction, bit counting, and rank scanning.
- Built the foundation for a move generator and minimax search using efficient attack masks and occupancy handling.

HTTP Server from Scratch (github.com/humankernel/http-server)

Apr 2025 — Apr 2025

- Built a minimal HTTP/1.1 server in Go from scratch using sockets without external frameworks.
- Implemented request parsing, routing, concurrency handling, and static file serving.
- Added proper response headers, MIME type detection, and error handling.
- Gained deep understanding of TCP networking, low-level protocols, and server architecture.

Terminal Chat (github.com/humankernel/chat-go)

Apr 2025 — Apr 2025

- Developed a real-time chat system in Go using HTTP/1.1 and concurrency handling.
- Implemented client-server communication, session management, and message delivery.
- Integrated error handling and support for multiple simultaneous users.
- Deepened knowledge of networking protocols and concurrent process synchronization.

SKILLS

- **Programming Languages:** C, Python, TypeScript, Go, HTML/CSS, SQL (PostgreSQL)
- **Technologies:** Linux, Git, Docker, React, TanStack Query, Astro, NextJS, Tailwind CSS, Pandas, NumPy
- **Languages:** Spanish (native), English (B1 – intermediate)